

Position paper of BIO Deutschland

From research to commercialisation: expanding technology transfer and translation in Germany

Berlin, 17 December 2024

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1. Introduction

This position paper sets out the current key demands of the technology transfer experts of the biotechnology industry association BIO Deutschland for how to improve the translation of scientific findings into practice. The biotechnology industry is highly innovative, entrepreneur driven, and capital intensive. Technology transfer offices at universities and research institutions play an indispensable role in the commercialisation of research results. Many different stakeholders in Germany can influence the success of technology transfer, and it is imperative that they do so. This position paper therefore makes recommendations for action by government (federal and state), academia, and industry.

2. Key demands and recommendations for action to promote technology transfer

2.1. Strengthen the innovation and entrepreneurial culture at universities and research institutions

- Foster cultural transformation towards an entrepreneurial mindset and more risk-taking among academic researchers.
- Promote dual career paths and cooperation between academia and industry.
- Simplify secondary employment permits and leaves of absence for commercial activities of academic researchers.

2.2. Establish and bolster technology transfer as a "third mission" at universities and research institutions

- Increase budget targets for basic and project funding of universities and research institutions so that an additional 5 percent is earmarked for both technology transfer and validation studies respectively.
- Allow exceptions to the ban on providing better employment conditions than comparable federal government employees (*Besserstellungsverbot*) in order to attract and retain professional and experienced technology transfer staff.
- Introduction of technology transfer indicators (such as the number of industry partnerships, of patents received and patent applications pending, of licenses and spin-outs, as well as the existence of incubators) to measure the quality of universities and research institutions.

2.3. Create the conditions that facilitate the financing and growth of spin-outs

- Maintain and expand existing support schemes for entrepreneurs.
- Make the tax and legal environment more attractive for venture capital funds and private investors.
- Enable pension funds and health insurance companies to invest more in biotechnology.

3. Rationale for the measures proposed to improve Germany as a location for translational and biotech innovation

3.1. Why the culture of innovation and entrepreneurship at universities and research institutions needs to be strengthened – and how to go about it

For many PhD students and postgraduate researchers at natural science faculties, the potential patenting and commercialisation of research results is not much of an issue, if one at all. Their first priority is usually to publish their findings at a high level. One reason for this is that the economic and entrepreneurial aspects of research and development (R&D) rarely feature in these faculties' curricula. However, there are exceptions to this norm, including in Germany at higher education institutions like TU Munich or TU Darmstadt. Programmes that promote entrepreneurial thinking and encourage students to leave the academic path (even if only temporarily) and translate novel research results into innovations should be a standard part of university curriculum, as they could bring about a change of perspective and direct more attention to commercialisation.

Yet simply increasing the amount of information available will not be enough to improve the innovation and start-up culture. Researchers often have a sceptical attitude towards private industry and in some cases considerable fear of contact. A switch to industry is seen as rather risky, even though an academic career is typically marked by a degree of uncertainty. In addition, it can be difficult or even impossible to move to industry, whether to a spin-out or to an established company, for a certain period of time and then return to academia. In order to enable more academic researchers to pursue this path, it must be made legally easier for them to take up (secondary) employment in industry. Bavaria, for instance, has now created a legal environment that does just that. This involves granting "a leave of absence to professors, generally for two semesters, to participate in commercial activities, including business start-ups, while retaining their salaries" (Article 61 (2) of the Bavarian Higher Education Innovation Act [BayHIGI]).

As mentioned above, a strong publication record is the most important factor in the career advancement of researchers. This measure of academic success, which has been established for decades, requires researchers to focus one-sidedly on publishing their findings in journals. There are a few universities and research institutions that already use industry partnerships or patents as criteria for making professorial appointments and hiring other staff. More universities should adopt this practice as it can help to create incentives for the commercial exploitation of research results.

Cooperation between industry and academia also exists in Germany of course. A successful example is the beLAB2122 BRIDGE, which is a partnership between the German Cancer Research Center (DKFZ), the European Molecular Biology Laboratory (EMBL), the Universities of Heidelberg, Frankfurt and Tübingen, and the companies Evotec and Bristol Myers Squibb. Partnerships such as these enable dual career paths between academia and industry and help build trust. Similar partnerships should be expanded and promoted.

3.2. Why the “third mission” at universities and research institutions needs to be established and bolstered – and how to go about it

At most of Germany's universities and some of its non-university research institutions, the commercial exploitation of research results is of secondary importance. The “third mission” of universities as a third pillar alongside research and teaching plays little or no role, and technology transfer offices are not well staffed or funded. The ban on providing better employment conditions than comparable federal government employees (*Besserstellungsverbot*) makes it impossible for the offices to attract and retain experienced staff with a background in industry. Germany's federal states should therefore place a much higher priority on the third mission at their universities and relax the aforementioned ban with respect to certain personnel decisions, in a similar way as under the German SPRIND Freedom Act. This is the only way to establish experienced and professional technology transfer offices at universities.

It takes considerable time and effort for technology transfer offices to prepare and successfully implement commercialisation activities. This requires the offices to have discussions with researchers to gather information and identify promising inventions, while also evaluating the discoveries for patentability and market potential. In the case of joint inventions, commercialisation agreements are concluded with the various partners. Once the evaluation process is complete, the technology transfer offices take steps to protect and market the inventions. This frequently involves submitting patent applications and supporting them in national and international venues until the patents are granted. Marketing activities include creating targeted materials, negotiating licensing and cooperation agreements, and assisting start-ups with such tasks as writing business plans, recruiting management personnel and acquiring investors. However, technology transfer offices in Germany rarely have sufficient budget to carry out these numerous essential activities.

In order to improve the chronic underfunding of technology transfer activities at universities, the basic budget as well as funding for relevant support schemes should be expanded so that at least an additional 5 percent is earmarked for both technology transfer infrastructure and validation studies respectively. The same should apply to project funding. Here, too, there should be an explicit budget for technology transfer. Part of this budget could be used for such purposes as fellowships aimed at facilitating exchange between technology transfer experts.

Although a few universities and research institutions already use patents and spin-outs as a measure of quality, the ranking and reputation of universities and research institutions in Germany is still largely determined by the number of publications they produce. It would therefore be helpful to introduce technology transfer indicators, so as to improve the translation of scientific discoveries into technological innovations. These could, for example, include the number of industry partnerships, the number of patents received and patent applications pending, and the number of licenses and spin-outs. This would put greater focus on translation as a measure of quality, as is already common practice in countries like the US and UK.

3.3. Why creating the right environment for the financing and growth of spin-outs is essential – and how to go about it

Government support schemes are available to entrepreneurs who need financial assistance to start their businesses. One such scheme is the EXIST programme of the German Federal Ministry for Economic Affairs and Climate Action (BMWK), which has become an important funding instrument for life science start-ups. Another support scheme called Gründungsoffensive Biotechnologie, or "GO-Bio" for short, was relaunched in spring 2024 after being suspended for six years. It is tailored specifically to the unique challenges of the biotech sector. The bioeconomy and biomedicine funding lines of the SME support scheme titled "KMU-innovativ" were also suspended for some time, but are currently open again. However, due to the uncertain budget situation since November 2024, there is concern that they could be suspended again in the near future. Yet it is essential for start-ups that such support schemes are stable and reliable. The programmes mentioned above must therefore be maintained as a matter of urgency.

The German Federal Agency for Breakthrough Innovation (SPRIND) supports start-ups based on promising disruptive inventions. The SPRIND Freedom Act has significantly bolstered the ability of the agency to carry out its mandate. The German Agency for Transfer and Innovation (DATI), which is a product of the 2021 coalition agreement, is still in the process of being founded. The agency is supposed to promote so-called innovation communities at the regional level, but it is not yet known what its final structure and objectives will be – except that it will be oriented towards excellence and has lost its original focus on universities of applied sciences. It is therefore not yet clear whether this support strategy will be a sensible one for DATI to undertake. However, in principle, BIO Deutschland advocates supporting established and successful structures and processes rather than establishing new agencies or structures.

After completing the founding process and securing early-stage financing, many biotech start-ups soon urgently need follow-up and growth financing and are faced with the question of how to raise this capital. In contrast to start-ups in the service sector, R&D in biotechnology is cost intensive, risky, and time consuming. Debt capital (loan financing) is generally not a financing option for start-ups in both the development of therapeutics and industrial (white) biotechnology. In the past few years, policymakers have become aware of this and recognised the need to take action.

Consequently, there has recently been a number of laws and pronouncements that acknowledge this need to facilitate equity financing for start-ups, with this funding coming from venture capital firms, business angels or family offices. Thanks to the HTGF Opportunity Fund launched in June 2024, the BMWK is now supporting young and growing companies through the HTGF portfolio.

Under the WIN initiative, launched in September 2024, businesses, industry associations, policymakers and the development bank KfW have formed an alliance committed to establishing Germany as leading financing hub and to promoting start-ups, innovation and venture capital in the country. A package of measures was agreed to improve the environment for innovation and growth capital in Germany. At the same time, the participating businesses have pledged to invest some €12 billion by 2030 to further strengthen the German venture capital ecosystem. The initiative's 10-point programme of measures sets out key steps for mobilising growth capital – for example, making it easier for small insurers and pension funds to invest in high-yield and high-risk asset classes such as venture capital, establishing innovation and growth capital as an asset class for high-net-worth individuals and other private investors, and improving the tax framework for investments in innovation and growth capital. In countries like the US, more venture capital is available because pension funds and insurance companies can invest in riskier asset classes such as biotech. In France, investments by private individuals in high-tech assets are tax privileged. The WIN initiative also aims to further improve the environment for IPOs, because whether an investor invests in a start-up is influenced by whether there are opportunities for a later exit through the stock market. In order to create a functioning financial market ecosystem in Europe, IPOs must be available as an exit channel, with young companies going public not only in the US, but also here. The WIN initiative should definitely be pursued further, as it contains very important financing options for biotech companies.

This position paper was prepared by BIO Deutschland's Working Group on Technology Transfer..

As Germany's national biotechnology industry association, BIO Deutschland has set itself the objective of supporting and promoting the development of an innovative economic sector based on modern biosciences. The association currently has over 390 members. Oliver Schacht, Ph.D., is Chairman of the Board of BIO Deutschland.

BIO Deutschland's supporting members and industry partners are AGC Biologics, Avia, AWS, Baker Tilly, Bayer, BioSpring, Boehringer Ingelheim, Centogene, Clariant, CMS Hasche Sigle, Deutsche Bank, Ernst & Young, Evotec, Janssen-Cilag, KPMG, Lonza, Merck, Miltenyi Biotec, MorphoSys, Novartis, Pfizer, PricewaterhouseCoopers, QIAGEN, Rentschler Biopharma, Roche Diagnostics, Sanofi Aventis Deutschland, SAP, Schmidt Versicherings Treuhand AG, Simmons & Simmons, Springer Nature, Thermo Fisher Scientific, TVM Life Sciences Management, Vertex Pharmaceuticals and ZETA.

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